# why do some groups experience greater mobility than others?

an IGE-based conditional-distribution framework linking absolute and relative intergenerational mobility, with origin-specific uncertainty

mauricio bucca

puc chile

# the problem

# context

- social mobility is a central theme in public discussions about fairness, opportunity, and the "rules of the game."
- framed—alongside wealth, poverty, and inequality—as a defining marker of a society's character (e.g., "the land of opportunity").
- commonly invoked as a public good: metaphors like "the social elevator" or a "level playing field" signal that rising should be attainable for all.

# the problem

- social mobility means different things in public debate and academic research
- two core concepts are often conflated:
  - **relative mobility:** how tightly children's positions track their parents'
  - **absolute mobility:** where children end up compared with their parents (in levels)

- so what counts as **"good" or "bad" mobility**? seems straightforward enough…
  - weaker parent-child associations → more "equality of opportunity"
  - more people surpassing their parents → more upward mobility

- **but not so fast!**: these two mobility types can move in opposite directions and have




"We want to create a society in which the son of a bus driver can go on not only to run but own the bus company."




"We want to create a society in which the son of a bus driver can go on not only to run but own the bus company."

**two ways this story could be true:**

1. the bus driver's son ends up owning the bus company, while the doctor's son winds up driving the bus *(rank reversal)*
2. the bus driver's son owns the bus company, and the doctor's son owns the clinic *(upward mobility for both)*

# this paper

**core claim:** properly understanding mobility requires examining both types simultaneously.

this paper introduces a **simple unified framework** that allows for that

# analytical framework

# definitions

**Relative mobility** captures how tightly children's economic standing tracks their parents'.

- Represented by the conditional distribution
  $f(y \mid x)$, where $x$ is parental income and $y$ is child income.

- the standard model is the **IGE workhorse**: $E[y \mid x] = \alpha + \beta x$

  where $x$ is parents' permanent log income and $y$ is children's log income.
  the slope $\beta$ is the **intergenerational elasticity (IGE)**, typically in the [0.2–0.6] range.

- population-level concept: **non-directional** and essentially a **zero-sum** measure of positional movement.

# definitions

**absolute mobility** compares children's income to their parents on a **level scale**, not a rank or positional scale.

- with a meaningful threshold $\delta$, classify mobility using the gap $y - x$:
  - **upward mobility:** $y > x + \delta$
  - **downward mobility:** $y < x - \delta$
  - **immobility:** $x - \delta \leq y \leq x + \delta$

- $\delta$ is a modeling choice, not estimated: baseline $\delta = \log(1.15)$ (a $\pm 15\%$ income band); results robust across $\tau \in [0, 0.30]$

# linking relative and absolute mobility

standard IGE specification:

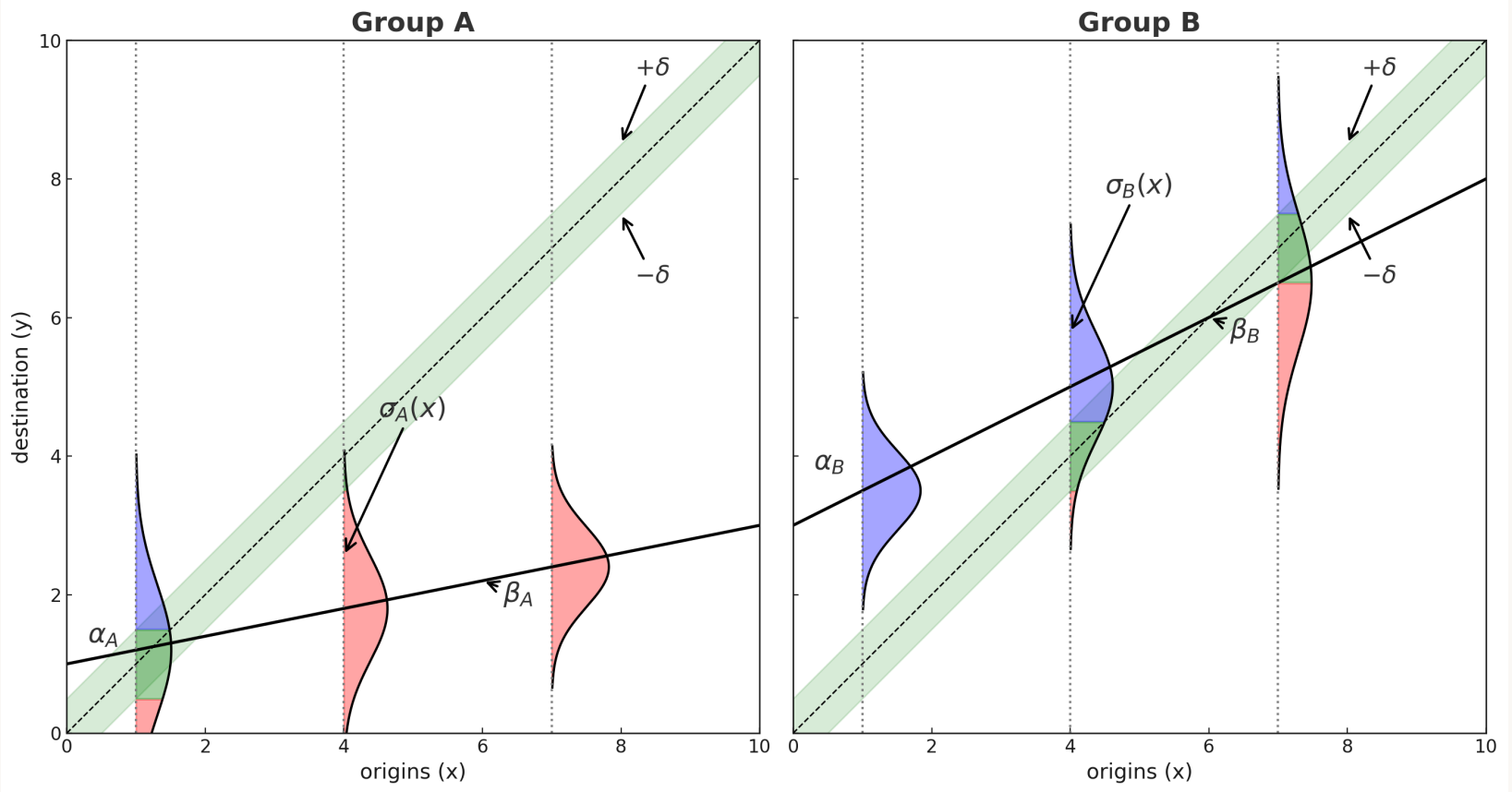
$$y_i \mid x_i \sim \text{Normal}\big(\mu_i = \alpha + \beta x_i,\ \sigma\big)$$

four key components:

| parameter | interpretation |
|---|---|
| $\alpha$ | income floor — expected income at a low **reference origin** $c$ (not literally the bottom) |
| $\beta$ | persistence (IGE) — degree of relative immobility |
| $\sigma$ | uncertainty — unpredictability in intergenerational transmission |
| $x$ | parental income — the baseline for absolute comparison |

*note: can be extended so both expected income $\mu_i(x)$ and income uncertainty $\sigma(x)$ vary with parental income*

# linking relative and absolute mobility




absolute mobility probabilities:

# linking relative and absolute mobility

we can express absolute mobility probabilities as a function of the IGE components:

- step 1: start with the income generation model: $y \mid x \sim \text{Normal}\Big(\mu(x) = \alpha + \beta\widetilde{x},\ \sigma(x)\Big)$, with $\widetilde{x} = x - c$ centered at a low reference point $c$

- step 2: standardized mobility thresholds (comparison is still against **raw** $x$)

$$z^{\uparrow} = \frac{x + \delta - \mu(x)}{\sigma(x)}$$
$$z^{\downarrow} = \frac{x - \delta - \mu(x)}{\sigma(x)}$$

- step 3: express **absolute mobility** probabilities using standard normal CDF $\Phi(\cdot)$

$$\Pr(M^{\uparrow}) = 1 - \Phi(z^{\uparrow})$$
$$\Pr(M^{\downarrow}) = \Phi(z^{\downarrow})$$

# drivers of absolute mobility

# driver 1: income floor ($\alpha$)

**question:** how do intercept shifts (uniform income growth) affect absolute mobility?

analytical result:

$$\frac{\partial \Pr(M^{\uparrow})}{\partial \alpha} = \frac{\phi(z^{\uparrow})}{\sigma}, \qquad \frac{\partial \Pr(M^{\downarrow})}{\partial \alpha} = -\frac{\phi(z^{\downarrow})}{\sigma}$$

where $\phi(\cdot)$ is the standard normal PDF

key insights:

- raising $\alpha$ shifts the entire income distribution upward
- **increases** upward mobility, **decreases** downward mobility
- effect is **uniform** across the parental income distribution

# driver 2: relative (im)mobility ($\beta$)

**question:** how do changes in relative mobility affect absolute mobility?

analytical result:

$$\frac{\partial \Pr(M^{\uparrow})}{\partial \beta} = \frac{\widetilde{x} \cdot \phi(z^{\uparrow})}{\sigma}, \qquad \frac{\partial \Pr(M^{\downarrow})}{\partial \beta} = -\frac{\widetilde{x} \cdot \phi(z^{\downarrow})}{\sigma}$$

key insights:

- $\beta$ **tilts** the distribution around the reference origin, rather than shifting it uniformly like $\alpha$
- effect is **proportional to centered parental income** $\widetilde{x} = x - c$: **zero** at the reference origin ($x = c$), growing with distance from it
- above the reference origin, increasing $\beta$ (more persistence) **increases** upward mobility and **decreases** downward mobility
- larger conditional income uncertainty $\sigma$ dilutes the impact of $\beta$

# driver 3: income uncertainty ($\sigma$)

**question:** how does the "predictability" of children's income affect absolute mobility?

analytical result:

$$\frac{\partial \Pr(M^{\uparrow})}{\partial \sigma} = \frac{z^{\uparrow}\phi(z^{\uparrow})}{\sigma}, \qquad \frac{\partial \Pr(M^{\downarrow})}{\partial \sigma} = -\frac{z^{\downarrow}\phi(z^{\downarrow})}{\sigma}$$

key insights:

- effect depends on **distance to thresholds** ($z^{\uparrow}$, $z^{\downarrow}$)
- at the bottom: increases upward mobility (lottery-ticket effect)
- at the top: increases downward mobility (greater vulnerability)

# driver 4: parental income ($x$)

**question:** how do social origins (parental income) affect absolute mobility?

analytical result (fixed dispersion):

$$\frac{\partial \Pr(M^{\uparrow})}{\partial x} = \frac{\beta - 1}{\sigma}\phi(z^{\uparrow}), \qquad \frac{\partial \Pr(M^{\downarrow})}{\partial x} = \frac{1 - \beta}{\sigma}\phi(z^{\downarrow})$$

key insights:

- since empirically $0 < \beta < 1$ :
  - upward mobility **decreases** with parental income $x$
  - downward mobility **increases** with parental income $x$
- even under **fixed persistence**, mobility varies substantially by origin
- origin effects are structural, not just compositional

# driver 4 (continued): a second channel under heterosk

**question:** once uncertainty $\sigma(x)$ also varies with origin, what else moves with parental income?

total derivative (both channels at once):

$$\frac{d\Pr(M^{\uparrow})}{dx} = \phi(z^{\uparrow})\left[\frac{-(1-\beta)}{\sigma(x)} + \lambda z^{\uparrow}\right], \qquad \frac{d\Pr(M^{\downarrow})}{dx} = \phi(z^{\downarrow})\left[\frac{1-\beta}{\sigma(x)} - \lambda z^{\downarrow}\right]$$

key insights:

- bracket #1 is **mean reversion** (driver 4, as before): pushes upward mobility down, downward mobility up at higher origins
- bracket #2 is a **new dispersion-gradient channel**, $\lambda z^{\uparrow}$ — absent when $\sigma$ is constant
- sign of $\lambda$ relative to $z$ decides whether the two channels **reinforce or offset**:
  - $\lambda < 0$ (narrowing uncertainty at the top) **reinforces** mean reversion

# counterfactual decomposition

# the parameter-swap counterfactual

**goal:** isolate the marginal contribution of each structural parameter to group differences

**challenge:** the effect of any single parameter depends on the levels of all others

**solution:** replace one parameter with its counterfactual value while holding others fixed. For groups $g$ and $\bar{g}$, I do three swaps:

1. $\alpha$**-swap:** plug in $\alpha_{\bar{g}}$ in place of $\alpha_g$
2. $\beta$**-swap:** plug in $\beta_{\bar{g}}$ in place of $\beta_g$
3. $\sigma$**-swap:** plug in $\sigma_{\bar{g}}(x)$ in place of $\sigma_g(x)$

> ***example question:*** *"how much upward mobility would group A experience if, holding everything else constant, it had group B's level of relative mobility?"*

# aggregation and composition

population-level consequences depend on the **distribution of parental origins**, which may differ across groups.

**aggregation** — average change in probability of mobility type $k$ when swapping parameter $\theta$ for group $g$, holding distribution of $x$ fixed:

$$\overline{\Delta}_k^{(g,\theta)} = \int \Delta_k^{(g,\theta)}(x)\, f^{(g)}(x)\, dx$$

**composition** — change in probability of mobility type $k$ induced solely by swapping the parental income distribution:

$$\overline{\Delta}_k^{(g,x)} = \int p_k^{(g)}(x) \left[ f^{(\bar{g})}(x) - f^{(g)}(x) \right] dx$$

distinguishes structural transmission (via parameters) from distributional positioning (via origins)

# model and estimation

# statistical model

**heteroskedastic IGE model:**

$$
\begin{aligned}
y_i \mid g, x_i &\sim \text{Normal}\left(\mu_g(x_i),\ \sqrt{\eta^2_{ig}/T_{ig} + \sigma^2_g(x_i)}\ \right) \\
\mu_g(x_i) &= \alpha_g + \beta_g \widetilde{x}_i\,, \qquad \widetilde{x}_i = x_i - c \\
\log \sigma_g(x_i) &= \gamma_g + \lambda_g \widetilde{x}_i
\end{aligned}
$$

**bayesian estimation approach:**

- $\eta^2_{ig}/T_{ig}$: sampling noise from averaging $T_{ig}$ annual earnings into "permanent" earnings — enters estimation, not the mobility thresholds
- hierarchical structure with group-specific parameters, non-centered parameterization
- weakly informative priors: informative prior for $\beta$: centered on empirically plausible range

# empirical application

# data: PSID 1968–2019

**sample:** 4,342 Black and White sons and daughters

- birth cohorts 1944–1989
- at least 2 positive earnings observations, ages 25–60
- valid parental earnings anchor + positive survey weight
- models estimated **separately by gender**

**earnings measures:**

- labor earnings: wages, salaries, self-employment, bonuses/tips, inflation-adjusted
- permanent child earnings: individual mean of log earnings
- permanent parental earnings: max(mother, father) age-42 anchor, from a centered 7-wave

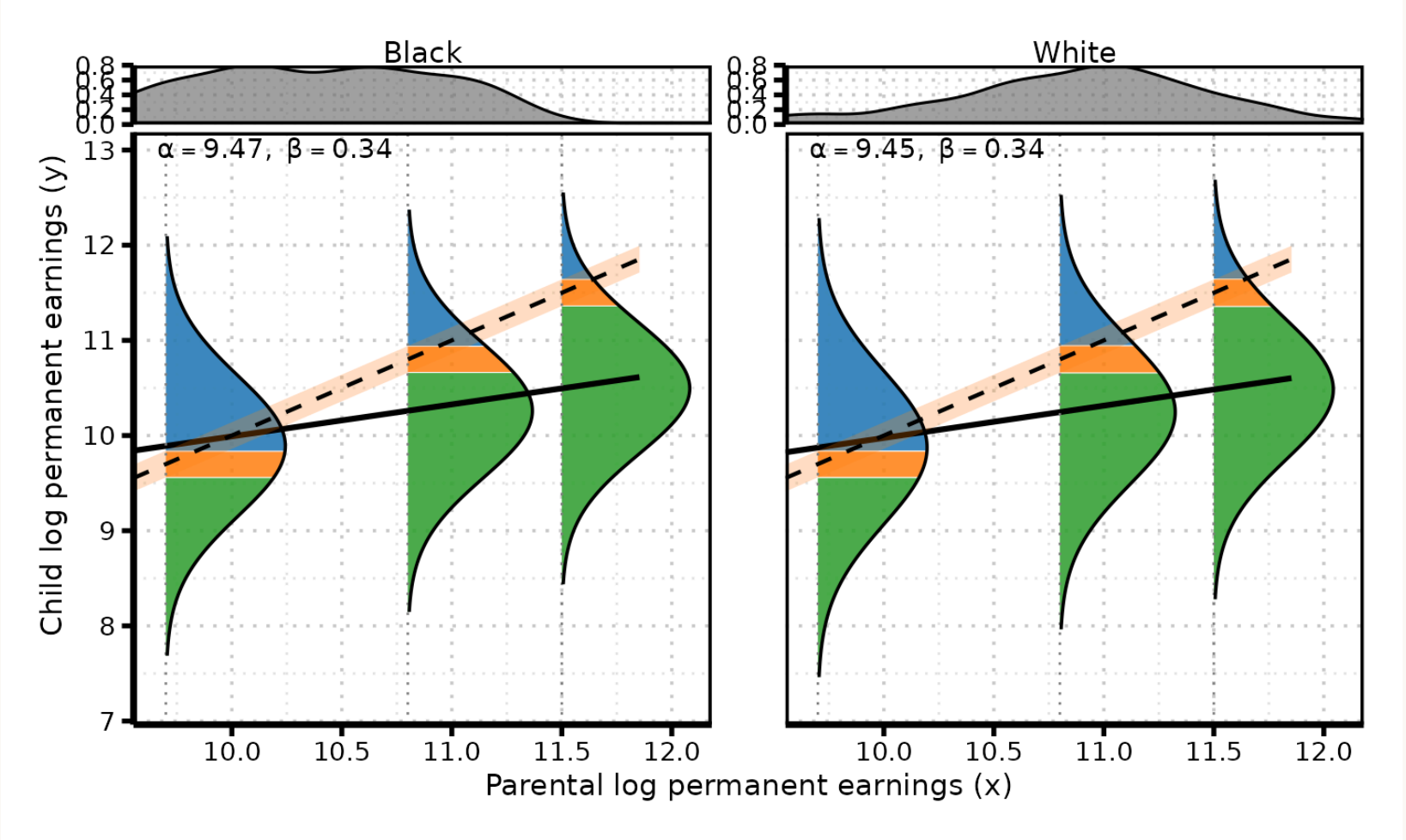
# estimated parameters, by gender

| parameter | black women | white women | black men | white men |
|---|---|---|---|---|
| $\alpha$ | 9.47 [9.28, 9.68] | 9.45 [9.33, 9.57] | 9.93 [9.66, 10.18] | 10.22 [10.12, 10.32] |
| $\beta$ | 0.34 [0.23, 0.43] | 0.34 [0.29, 0.39] | 0.28 [0.16, 0.41] | 0.26 [0.22, 0.29] |
| $\gamma$ | -0.26 [-0.51, -0.07] | -0.16 [-0.27, -0.04] | -0.36 [-0.56, -0.07] | -0.40 [-0.52, -0.28] |
| $\lambda$ | -0.04 [-0.13, 0.08] | -0.05 [-0.10, -0.00] | -0.12 [-0.27, -0.01] | -0.06 [-0.10, -0.01] |

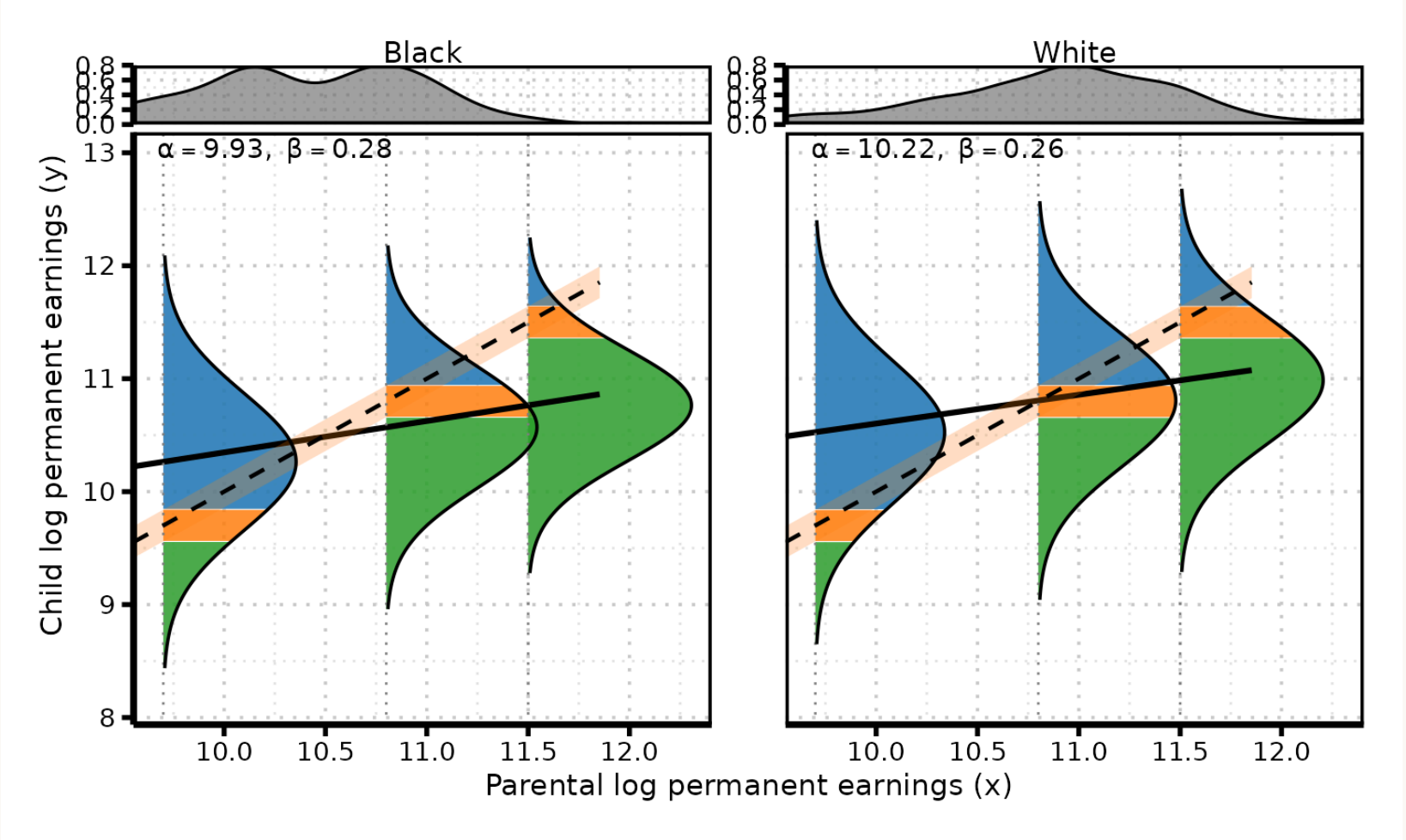
**key patterns:**

- **women:** floors, persistence, and dispersion are nearly identical across race — conditional distributions essentially overlap.
- **men:** persistence is similar ($\beta \approx 0.28$ vs. $0.26$), but White men's **earnings floor is ~0.29 log points higher** ($\approx 34\%$ more earnings at every origin)

# mobility profiles: women



# mobility profiles: men



# substantive interpretation

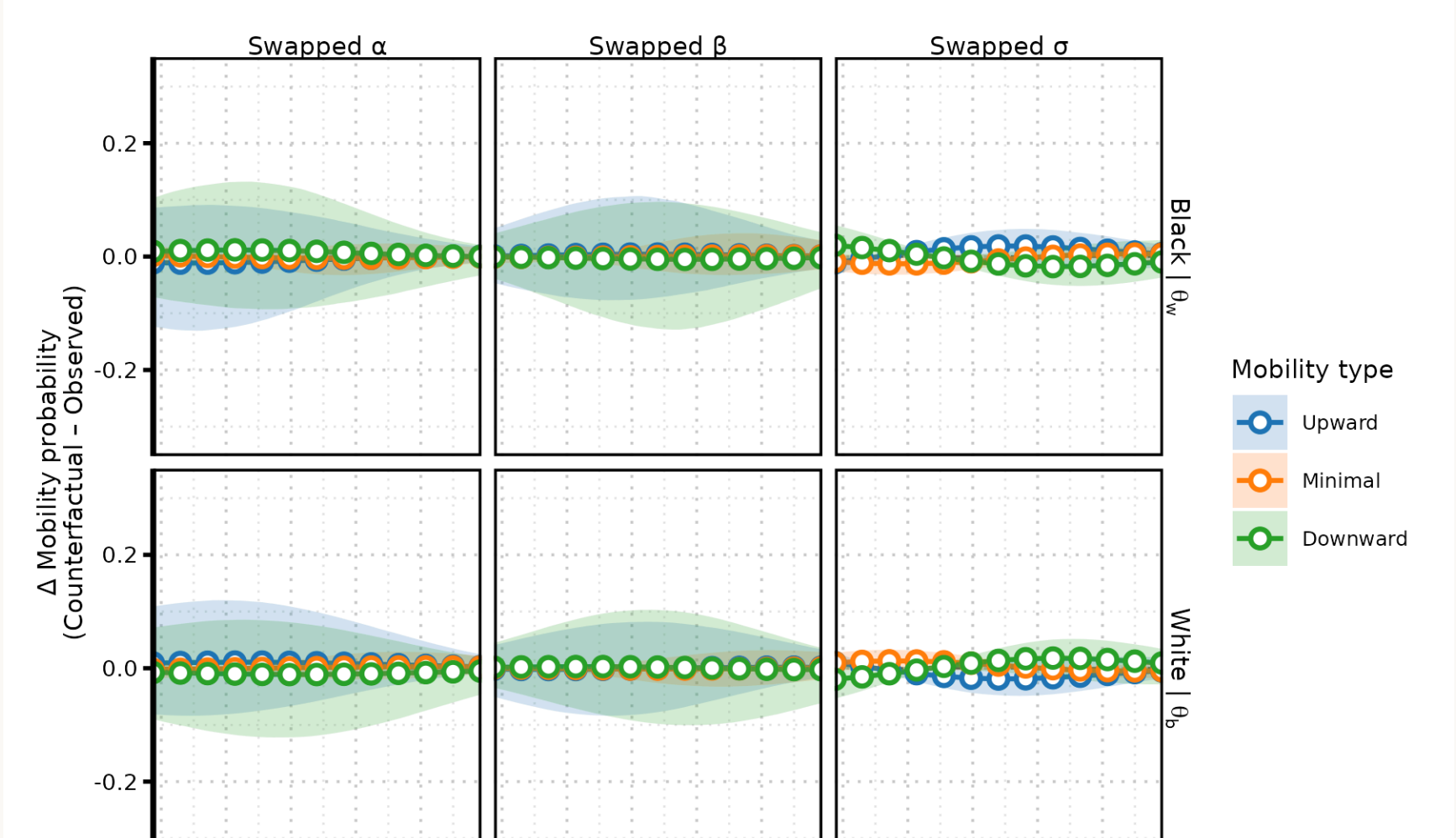
### women: nearly indistinguishable conditional processes

- same floor, same persistence, same dispersion — Black and White daughters with the same parental earnings are predicted to earn about the same.
- whatever racial gap exists must come from **where** families sit in the parental-earnings distribution, not from how earnings are transmitted.

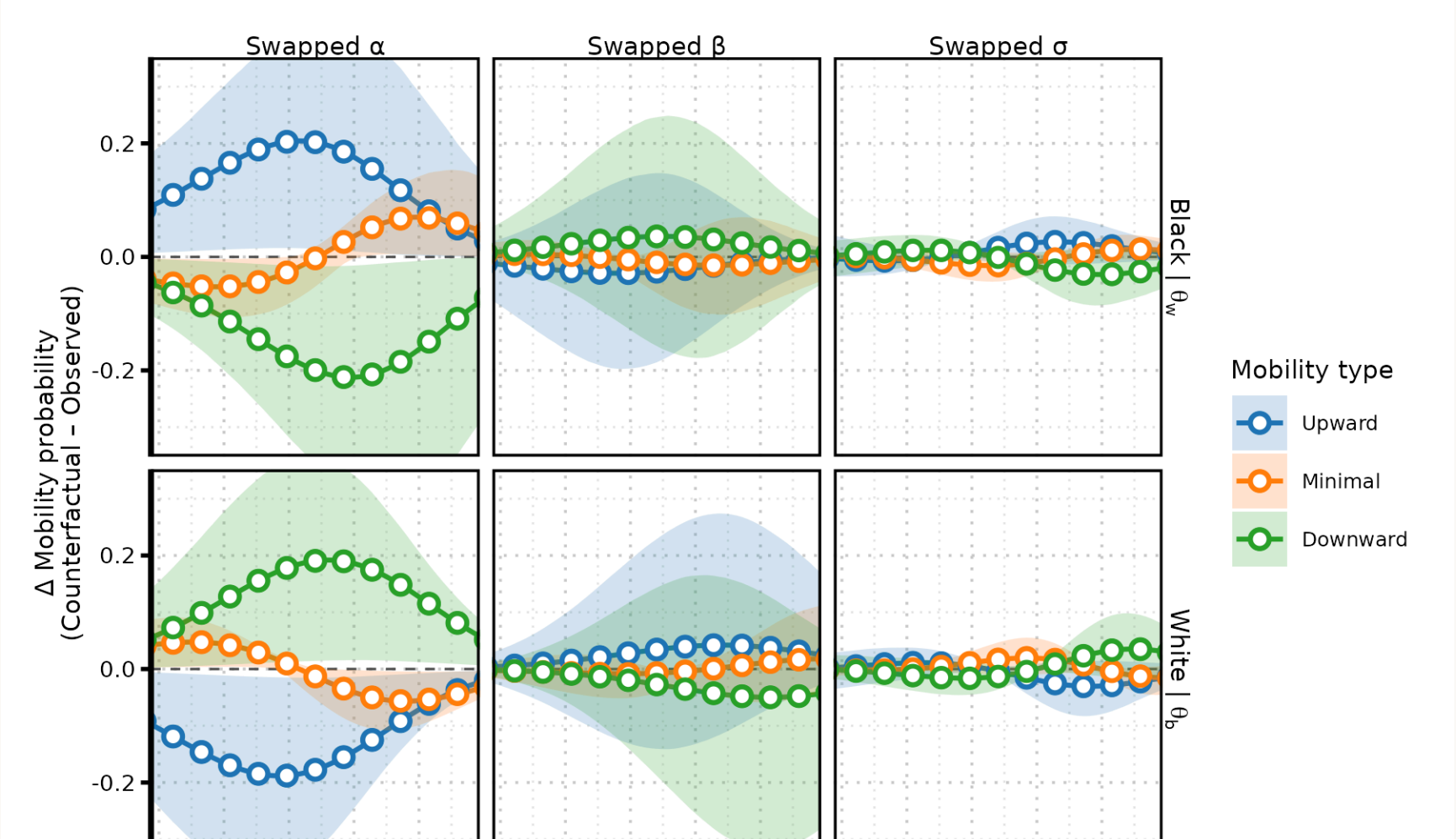
### men: a floor gap, not a persistence gap

- White sons start from a materially higher conditional mean at every origin — **not** because advantages transmit more strongly, but because the reference-origin floor itself is higher.
- same mean-reversion dynamics (similar $\beta$, similar $\lambda$) apply on top of that shifted floor.

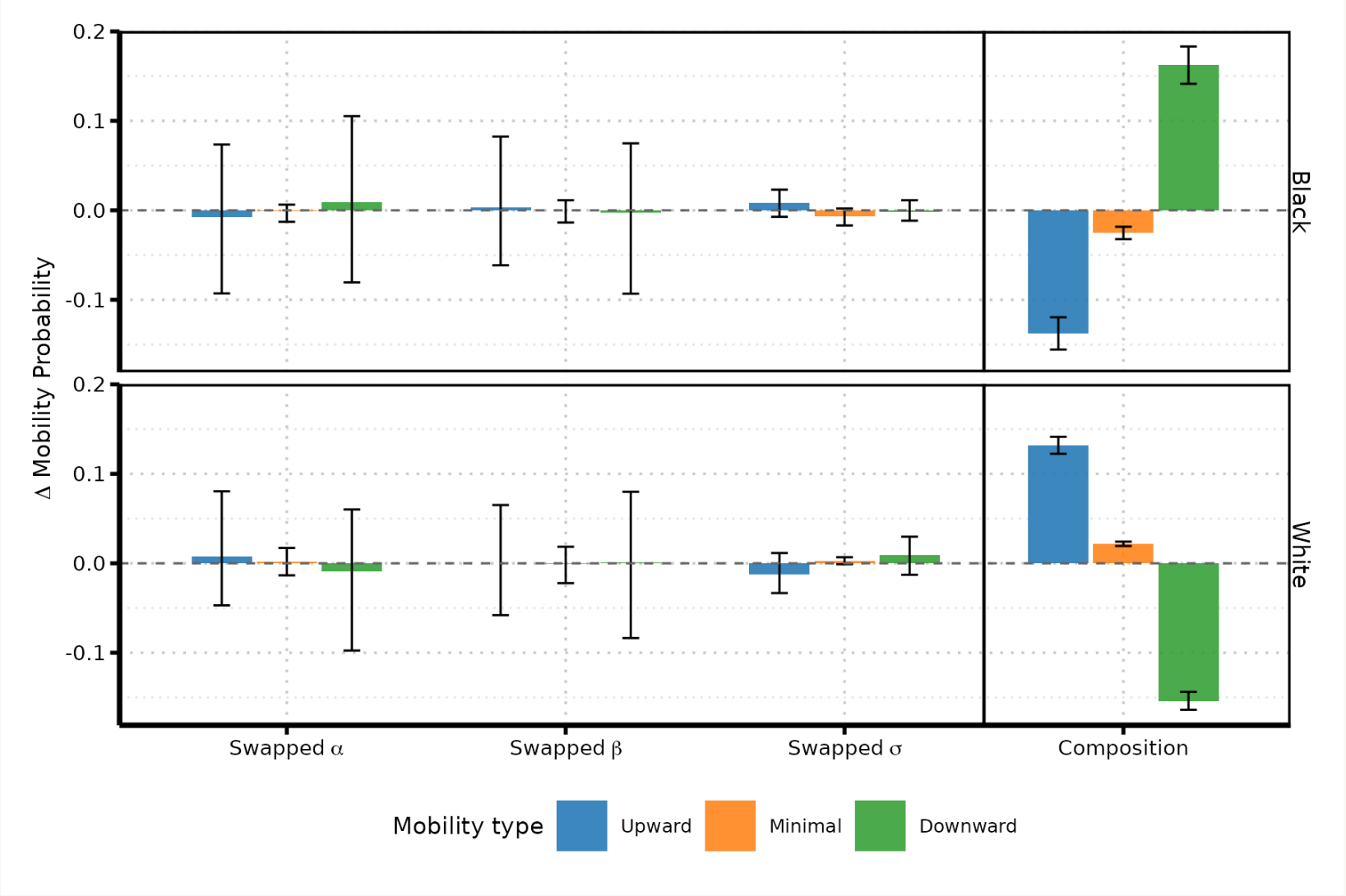
# pointwise counterfactual effects: women



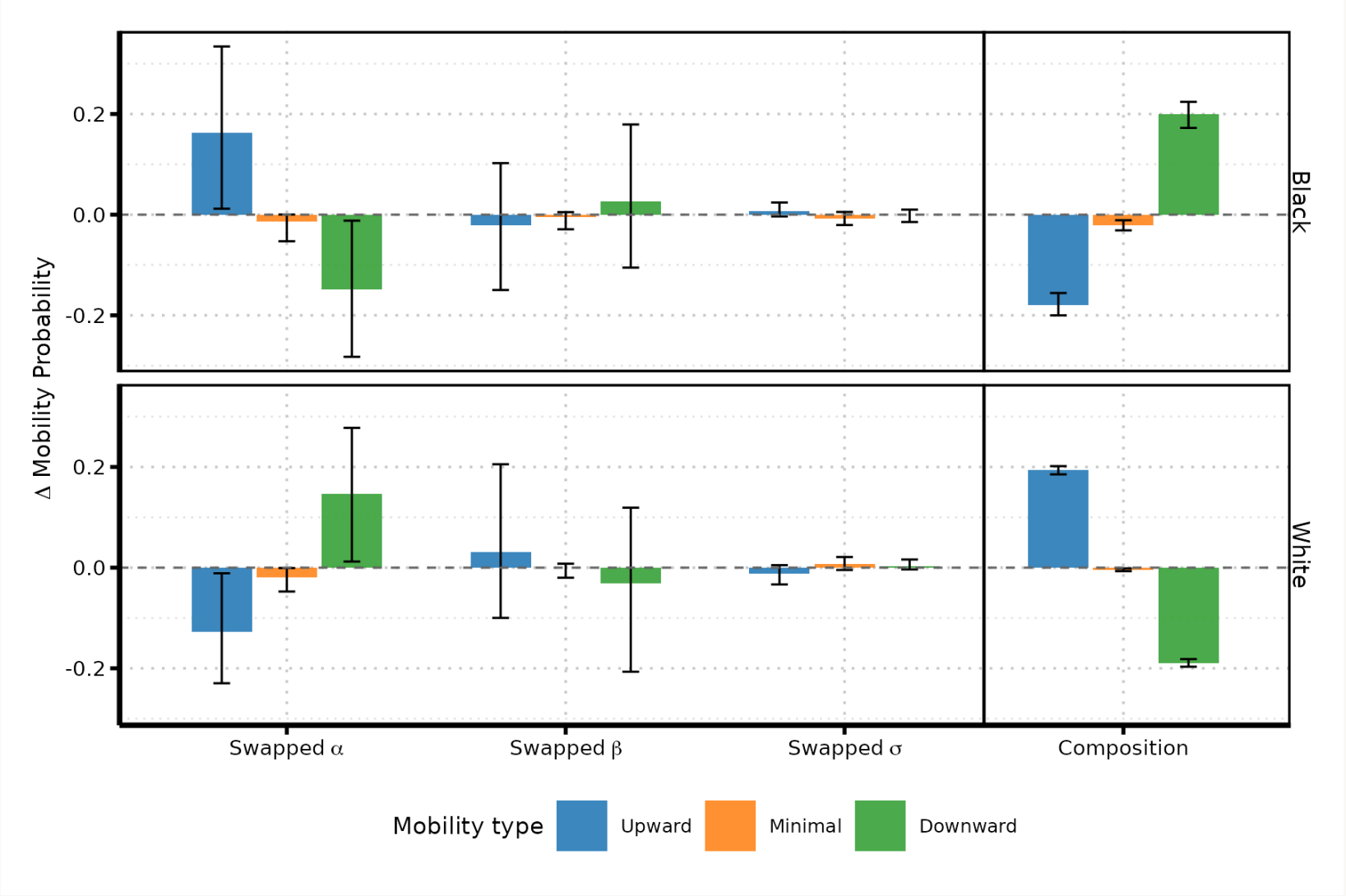
# pointwise counterfactual effects: men



# aggregation and composition: women



# aggregation and composition: men



# robustness: selection into positive earnings

- estimates condition on **positive earnings** — selection into that state is not origin-neutral.
- men: retention roughly flat by origin for both races.
- women: diverges by origin, in **opposite directions** — Black women's earnings-retention rises with origin ($\approx 0.85 \rightarrow 0.94$); White women's falls ($\approx 0.95 \rightarrow 0.89$, plausibly household specialization).
- imposing a hypothetical zero-earnings share ($q$ up to 15%) mainly shifts levels at **lower** origins; counterfactual signs/ordering unchanged, just scaled by $(1 - q)$.

# robustness: equivalized family income

- replacing own earnings with **equivalized family income** (less sensitive to labor-force attachment) preserves the compositional story overall…
- …but changes the picture for **women**: a **0.42 log-point** White–Black floor gap emerges that is absent in the earnings analysis.
- under family income, women now show the **same opposing-forces pattern as men**: floor disadvantage vs. compositional advantage.
- conclusion: the compositional finding is robust across outcomes; which **conditional** parameter matters depends on whether mobility is measured in own earnings or pooled family resources.

# key findings

**same aggregate summary — "composition dominates" — describes two different situations:**

> - ***women:*** *virtually no conditional gap → composition explains almost all of the observed (pro-Black) upward-mobility gap*
> - ***men:*** *a real earnings-floor disadvantage for Black men is* ***masked*** *by an offsetting compositional advantage from poorer origins*

disentangling conditional (floor/persistence/dispersion) from compositional (origins) margins — separately by gender — is what makes this visible.

# contributions and implications

- provides a unified, IGE-based conditional-distribution framework linking absolute and relative mobility, extended to allow **origin-dependent uncertainty**.
- shows uncertainty can **reinforce or offset** mean reversion — a second channel beyond persistence through which origin shapes absolute mobility.
- offers non-additive, one-factor-swap counterfactuals to disentangle floor, persistence, dispersion, and **compositional** (origin-distribution) contributions to group gaps.
- applied to Black–White earnings mobility: the **same aggregate gap** can hide a real conditional disadvantage (men) or reflect almost no conditional gap at all (women) — visible only once conditional and compositional margins are separated, by gender.

# thank you

mauricio bucca — puc chile · mebucca@uc.cl

**all replication materials available**

# appendix

# monte carlo validation

it works! trust me.

concretely: shrinking dispersion in Group A moves its implied correlation $\rho$ from $\approx 0.77$ to $\approx 0.97$; widening dispersion in Group B moves its $\rho$ from $\approx 0.89$ down to $\approx 0.52$ — with upward/downward mobility risk shifting accordingly, recovering the known data-generating process.